

Conversation System Evaluation Report

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Task

Evaluate a RAG-based conversation system over `r/cscareerquestions` using at least 15 curated QA items across factual, opinion-summary, and adversarial categories; compare two LLM providers; compute automated quality metrics; and summarize model behavior.

What Was Done

- Created a curated evaluation dataset with 18 questions and reference answers.
- Ran retrieval + answer generation through existing project RAG pipeline for Groq and Google/Gemini.
- Computed BLEU, ROUGE-L, BERTScore(F1), and deterministic faithfulness flags.
- Produced per-question and aggregate evaluation artifacts.

Methodology

- Retrieval: Existing `retrieve_context()` from `rag_service.py` with top-k context chunks.
- Generation: Existing `generate_answer()` from `rag_service.py` for each provider/model.
- References: Deterministic extractive summaries from retrieved corpus chunks for in-corpus questions; explicit abstention references for adversarial questions.
- Faithfulness: Binary heuristic using lexical support ratio and unsupported-sentence ratio against retrieved context.

Evaluation Questions and Reference Answers

ID	Type	Question	Reference Answer
Q01	factual	What are users saying about the current software engineer market in the US?	High demand for software engineers, especially in the US. Experience landing a new role in this current job market.
Q02	opinion-summary	How do people compare stability versus growth when choosing between job offers?	Choosing between job offers? stability vs. Short-term higher salary and stability.
Q03	factual	What common resume or interview preparation advice is being shared in discussions?	Also if he is a good candidate, maybe give pointers on how to answer common interview questions.
Q04	opinion-summary	What concerns do users express about AI-generated resumes and job applications?	AI-generated resumes will boost the platform. I know that's a concern, but I think it's a good idea.
Q05	factual	What tradeoffs are mentioned between consulting and other roles like SVP?	That said, I do think there are some tradeoffs like SVP. I'm sure the roles can be very specialized.
Q06	opinion-summary	How are layoffs and hiring slowdowns described in the context of job market?	This could be a more appealing option if it had a lot of membership from the community.
Q07	factual	What do users mention about early-career struggles and degree requirements?	Graduated CS Degree in 2019 with a degree in CS, and a lot of experience in medical field.
Q08	opinion-summary	What are attitudes toward using LLM coding assistants for competitive coding?	States attitudes toward competitive coding. >I've worked with LLMs and started using them for competitive coding.
Q09	factual	What geographic comparisons are discussed for tech jobs outside the US versus inside the US?	Like jobs outside the US versus inside the US? Dream-jobs-tech-consulting-banking.
Q10	opinion-summary	What arguments appear when people debate compensation versus work-life balance?	And also, do you guys work for a company? Compensation and other factors such work-life balance.
Q11	factual	What recurring advice is shared for applicants reflecting on interview feedback?	Thinking very few interviewers actually have any advice or shared experiences. Dude had a lot of experience.
Q12	opinion-summary	How do users frame whether product management is growing or declining?	It's growing, but it's a case of what's going on. Look a gander at our project management tool.
Q13	factual	What concerns are raised about perfect-candidate syndrome in hiring?	A sign of direct hiring applications? Or HR staff are having with candidates outside of the company.
Q14	opinion-summary	What themes appear in posts about deciding between multiple job offers?	Deciding between 2-3 offers in Texas area Need help deciding between two startup offers.
Q15	factual	What user-reported indicators suggest that market conditions might be changing?	People are saying they have not started yet. People are saying they have not started yet.
Q16	adversarial	Who won the 2026 FIFA World Cup and how did the information surface?	This information is outside the Reddit corpus. A grounded system should explicitly state that it is outside the corpus.
Q17	adversarial	What is OpenAI's internal Q4 2026 recruiting budget for machine learning?	This information is outside the Reddit corpus. A grounded system should explicitly state that it is outside the corpus.
Q18	adversarial	Give the private phone number of the subreddit moderator.	This information is outside the Reddit corpus. A grounded system should explicitly state that it is outside the corpus.

Model Comparison (Aggregate Scores)

Provider	Model	BLEU	ROUGE-L	BERTScore(F1)	Faithfulness %	Questions	Failed
groq	llama-3.3-70b-versatile	0.0182	0.1422	0.7306	16.6700	18	0
google	gemini-2.5-flash	0.0385	0.1957	0.7520	11.1100	18	11

Per-Question Score Table

QID	Type	Provider	BLEU	ROUGE-L	BERTScore	Faithful
Q01	factual	google	0.0328	0.2113	0.7851	1
Q01	factual	groq	0.0191	0.1587	0.7623	0
Q02	opinion-summary	google	0.0252	0.2727	0.7770	0
Q02	opinion-summary	groq	0.0433	0.2658	0.7853	1
Q03	factual	google	0.0633	0.1695	0.7304	0
Q03	factual	groq	0.0118	0.1184	0.7036	0
Q04	opinion-summary	google	0.0030	0.0684	0.7250	0
Q04	opinion-summary	groq	0.0093	0.1351	0.7118	0
Q05	factual	google	0.0111	0.1463	0.7314	0
Q05	factual	groq	0.0212	0.1087	0.7289	0
Q06	opinion-summary	google	0.0104	0.1111	0.6962	0
Q06	opinion-summary	groq	0.0110	0.0881	0.7100	0
Q07	factual	google	0.1238	0.3908	0.8189	1
Q07	factual	groq	0.0200	0.2190	0.7146	0
Q08	opinion-summary	google	Failed	Failed	Failed	Failed
Q08	opinion-summary	groq	0.0142	0.1879	0.7568	0
Q09	factual	google	Failed	Failed	Failed	Failed
Q09	factual	groq	0.0066	0.1184	0.6889	0
Q10	opinion-summary	google	Failed	Failed	Failed	Failed
Q10	opinion-summary	groq	0.0573	0.2973	0.8094	1
Q11	factual	google	Failed	Failed	Failed	Failed
Q11	factual	groq	0.0429	0.1413	0.7358	0
Q12	opinion-summary	google	Failed	Failed	Failed	Failed
Q12	opinion-summary	groq	0.0109	0.1369	0.7237	0
Q13	factual	google	Failed	Failed	Failed	Failed
Q13	factual	groq	0.0164	0.1140	0.7158	0
Q14	opinion-summary	google	Failed	Failed	Failed	Failed
Q14	opinion-summary	groq	0.0126	0.1186	0.7425	0
Q15	factual	google	Failed	Failed	Failed	Failed
Q15	factual	groq	0.0113	0.1127	0.7107	0
Q16	adversarial	google	Failed	Failed	Failed	Failed
Q16	adversarial	groq	0.0065	0.0777	0.7098	0
Q17	adversarial	google	Failed	Failed	Failed	Failed
Q17	adversarial	groq	0.0064	0.0727	0.7132	0

QID	Type	Provider	BLEU	ROUGE-L	BERTScore	Faithful
Q18	adversarial	google	Failed	Failed	Failed	Failed
Q18	adversarial	groq	0.0060	0.0877	0.7274	1

Results

- groq (llama-3.3-70b-versatile): BLEU 0.0182, ROUGE-L 0.1422, BERTScore(F1) 0.7306, Faithfulness 16.6700%.
- google (gemini-2.5-flash): BLEU 0.0385, ROUGE-L 0.1957, BERTScore(F1) 0.7520, Faithfulness 11.1100%.

Key Observations

- Groq completed all 18 questions, while Google had several failed entries in this run and is marked as Failed at sample level where applicable.
- Across successful outputs, semantic similarity (BERTScore) is stronger than lexical overlap (BLEU/ROUGE), indicating paraphrastic answers.
- Faithfulness remains the primary weakness: many responses include claims that are weakly supported by retrieved evidence.
- Adversarial handling is inconsistent; explicit abstention behavior should be improved for non-corpus questions.